ComplianceKB: An Auditable Regulatory Knowledge Model for Human-Supervised Compliance Reasoning

Anonymous submission

Abstract

Regulatory compliance assistants must identify duties, explain when they apply, and specify what evidence a reviewer should inspect. We present the Regulatory Knowledge Model (RKM), which connects each obligation to applicability descriptors, source provenance, a workflow trigger, and an audit-evidence category. This representation provides a reproducible interface between regulatory interpretation, model output, and human verification. ComplianceKB v1.2.0 instantiates it with 108 records from four federal OSHA frameworks, reviewed against official sources through an AI-assisted author verification process. A separate blinded annotation exercise on 50 retained historical IDs yields Cohen's κ of 0.234 for risk, 0.642 for evidence, and 0.443 for trigger. In an exploratory evaluation of 24 operational scenarios and 240 model calls, structured RKM context improves trigger and evidence F1 over identical hybrid retrieval candidates for two language models. Requirement selection, citations, and decision accuracy do not consistently improve. BM25, MiniLM, E5, and rank-fusion comparisons provide broader retrieval references. We formalize the pipeline, provide a worked regulatory example, and distinguish source integrity, annotation reliability, operational-label recovery, and workplace compliance. The findings support an inspectable audit-support architecture while retaining human responsibility for legal interpretation, evidence sufficiency, and safety-critical decisions.

1 Introduction

An industrial compliance question rarely ends with finding a paragraph. When a plant introduces a hazardous chemical or changes a covered process, a reviewer must determine which duties apply, when action is required, and what records or observable conditions would support an audit. A language model can summarize the relevant text, but the resulting answer must still be tied to a specific source, a particular version, and an operational decision. Repeating this interpretation independently for every query also makes inconsistent labels and unrecorded assumptions difficult to detect.

Compliance research already offers rule languages, structured regulatory extraction, and retrieval-augmented generation (RAG) [1–6]. Our research question is therefore narrower than whether AI can understand regulations: can a reusable, source-linked obligation representation make operational compliance outputs easier to reproduce and inspect, and which judgments remain dependent on people? The distinction matters because a plausible answer, a source-supported dataset record, and a legally sufficient compliance determination are different objects of validation.

We study this question through RKM and its ComplianceKB implementation. RKM assigns an identity to each obligation and connects its wording to applicability descriptors, the cited source, an operational trigger, and an audit-evidence type. A training requirement can thereby become a reusable task specification: which source supports the duty, which event should bring it to attention, and what kind of training record a reviewer needs. The model output remains a proposed audit checklist; neither an evidence-type label nor a citation proves that a workplace has complied.

The contributions are threefold. First, we specify an audit-support knowledge architecture and a concrete transformation from regulatory text to reviewed records and traceable model outputs. Second, we provide a versioned case study that separates structural validation, source verification, annotation agreement, and downstream evaluation, with complete author source review and an independently completed second annotation. Third, we evaluate how curated operational fields change outputs under controlled candidate retrieval, and report both their benefits and the remaining review burden. These are methodological and empirical contributions to human-supervised compliance systems, rather than a new language model, a new retrieval algorithm, or a claim to have established an industry standard.

2 Related Work and Positioning

Business-process compliance research studies the relationship between formalized norms and process behavior [1]. LegalRuleML supports expressive legal rule representation, including deontic and temporal concerns [2]. These traditions provide richer formal semantics than our compact records. RKM instead targets the practical handoff between an interpreted duty and the evidence a human reviewer should inspect; it does not introduce a new legal logic or execute a complete compliance proof.

Regulatory information extraction also predates RKM. CODE-ACCORD annotates entities and relations in building regulations to support automated rule generation [3]. ContractNLI combines contract-level inference with supporting text spans [4]. These resources demonstrate that machine-readable rules and source-grounded explanations are established goals. Our additional emphasis is a reviewed obligation as the maintained unit, with operational labels and an explicit distinction between regulatory support and organizational evidence.

Recent RAG work is directly relevant. Sun et al. combine structured regulatory events, business-process information, and retrieval-based reasoning in a compliance-checking framework [5]. Gokhan and Briscoe study multi-passage regulatory QA with lexical, semantic, and graph-derived retrieval signals [6]. The RIRAG shared task also evaluates regulatory retrieval and answer generation [7]. Accordingly, we do not claim that event-oriented representation, regulatory RAG, or compliance agents are new. Our comparison asks what happens when curated applicability, trigger, evidence, and provenance fields are exposed alongside the same candidate excerpts. It measures the operational usefulness of this representation under a fixed corpus, rather than superiority over the systems in [5–7] on their datasets.

Table 1: Relationship to prior work; distinctions concern the study focus, not absence of features in other systems.

Prior work	Primary focus	Focus of this study
Compliance formalisms [1, 2]	Norm and process semantics	Compact obligation-to-audit handoff
CODE-ACCORD [3]	Regulatory entity/relation annotation	Maintained duties, triggers, and evidence
ContractNLI [4]	Inference with supporting text	Source support plus workplace-evidence type
Regulatory RAG [5–7]	Retrieval and regulatory answering	Paired comparison of operational context

The human-review boundary connects this work to selective classification, where coverage is traded against prediction risk [8]. Our prototype offers missing-information and escalation decisions, but it has no fitted confidence threshold or calibrated probability of error. We use this literature to state the evaluation needed for future review-allocation policies, not to attribute statistical guarantees to the current agent.

3 Regulatory Knowledge Model

3.1 Representation and audit semantics

Let a regulatory record r_i be represented by the following tuple, whose elements correspond to fields in the released JSON schema:

$$r_i = (\mathrm{id}_i, u_i, m_i, A_i, t_i, e_i, v_i, P_i). \tag{1}$$

Here u_i is the obligation restatement; m_i contains mandatory status, requirement type, and an ordinal risk annotation; A_i contains applicability descriptors; t_i is the primary workflow trigger; e_i is the primary evidence type; and v_i is the verification method. Provenance P_i contains the source clause, URL, excerpt, source version, retrieval and review metadata, content hash, and curation status. These fields separate the duty, its operational interpretation, and the material used to justify it.

The design aims at one independently auditable duty per record. Atomicity is an editorial rule enforced through review, not a mechanically proved property: a provision may contain several conditions or related duties. Additional triggers and supporting clauses are retained for some records, but the benchmark exposes only the primary trigger and the fields listed in Algorithm 1. Applicability fields are computable descriptors, not a complete legal predicate language. In particular, industry codes cannot establish PSM coverage without the substance, quantity, process, and exception conditions in the regulation. The evaluated agent does not hard-filter candidates by NAICS or treat these descriptors as binding legal conclusions.

The audit-evidence field names a principal artifact or observable condition, such as a training record, permit, procedure, or physical condition. It does not contain that workplace artifact. This makes two evidence relationships explicit: the official excerpt supports the regulatory duty, while a separately collected workplace record may support performance of the duty. Confusing these relationships would let a system mistake citing a rule for satisfying it.

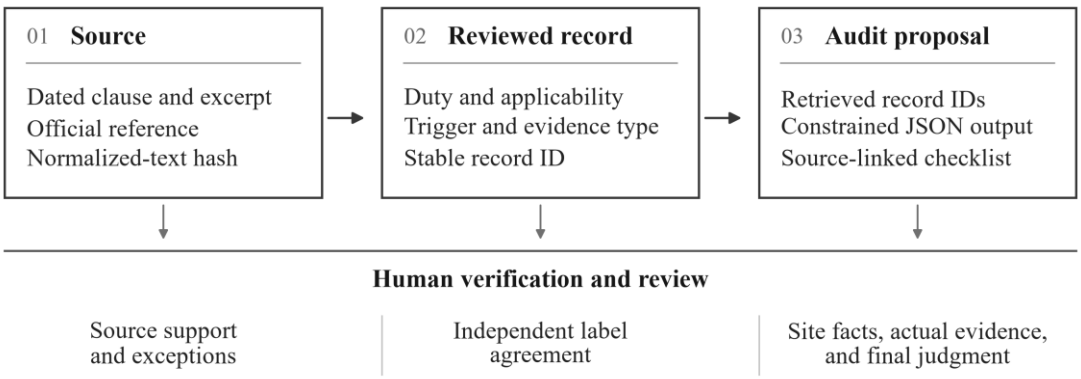


Figure 1: Source provenance, reviewed interpretation, and generated audit proposals remain separate objects of human verification.

3.2 Worked transformation from source to record

Consider OSHA 29 CFR 1910.119(g)(3), which concerns documentation of process-operator training [15]. The source includes: "The employer shall prepare a record which contains the identity of the employee, the date of training, and the means used to verify that the employee understood the training." This source passage identifies an artifact and its required contents, which are more useful for an audit than a generic instruction to ensure training.

Table 2: A source-to-record example from the frozen experimental release.

Field	Stored content / interpretation
Record and source	OSHA-PSM-018; 29 CFR 1910.119(g)(3)
Requirement	Ascertain and document that each process employee received and understood the required training.
Source-required record contents	Employee identity; training date; means used to verify understanding.
Evidence / method	training_record / record_review
Primary / additional trigger	onboarding_new_employee / training_refresh
Integrity reference	Normalized excerpt SHA-256 begins 9d3264fe53abd1d4; full digest is retained in the JSON.
Human judgment still needed	Verify the record contents, training scope, and actual understanding; inspect applicability and exceptions.

The table is a projection of the released record, not a reconstructed extraction prompt. The complete excerpt and its normalized-text hash remain in the dataset. A reviewer can trace the paraphrase to the paragraph and determine which details must be checked in an actual training record. This trace also reveals representational limits: the primary trigger stored for this record is onboarding_ new_employee, whereas the rule also concerns refresher training. The release preserves training_

refresh as an additional trigger, but the benchmark omits this extra field. RKM therefore makes an interpretation inspectable without making every interpretation unambiguous.

3.3 Source integrity and version changes

The released record hash is SHA-256 of a normalized excerpt, not of the entire live webpage. The exact normalization used in the build is shown below. It lowercases text, collapses whitespace, and removes characters outside the displayed pattern before hashing UTF-8 bytes.

```
def source_hash(text):
    text = re.sub(r"\s+", " ", text.lower().strip())
    text = re.sub(r"[^\w\s§.()\-]", "", text)
    return hashlib.sha256(text.encode("utf-8")).hexdigest()
```

Recomputing this value detects changes to the normalized stored excerpt. It does not prove that the excerpt was authoritative, complete, or legally interpreted correctly, and normalization intentionally ignores some typographic differences. A separate checksum of the released file protects exact bytes. To assess source drift, one must acquire the same provision at two dated snapshots and compare the texts before mapping changed clauses to affected record IDs.

The saved version audit compared 14 sections cited by the original 89 records between 2024-07-01 and 2026-09-24. It reported one changed section, HazCom [14], with changed cited paragraphs affecting three existing records; the targeted review preserved the historical baseline. The experimental release consistently declares source_version=2024-07-01, although its URLs point to live /current/ pages. Readers must use the stored excerpts and historical date when reproducing the study. The retained comparison report supports a change-review workflow; the original downloaded XML files are not in the experimental archive, so the snapshot comparison cannot be replayed offline from that archive alone.

Hashes and dated references could support a future graph linking a clause snapshot to its obligations and dependent audit tasks. They do not themselves provide amendment semantics, effective-date intervals, conflict resolution, or temporal reasoning. Those capabilities require explicit version relations and legal review beyond the present implementation.

4 Construction and Human Verification

4.1 Data lineage and release scope

The initial 89-record resource was assembled from selected operational provisions in PSM, HazCom, Recordkeeping, and Lockout/Tagout (LOTO). AI assistance supported source segmentation, candidate obligation restatement, classification, applicability mapping, operational annotation, and code generation. The retained historical log contains prompt summaries and Claude-family model descriptions, but no complete per-run extraction requests. Exact historical model IDs and inference parameters cannot be recovered reliably from that log.

The revision uses a non-destructive correction process. A full review of the 89 records produced corrections and exposed 19 additional or separated duties. Those 19 received a targeted review before being included in v1.2.0, producing the fixed 108-record experimental resource. The build applies recorded corrections and generates consistent JSON, CSV, JSON-LD, and SQL exports. A later coverage review produced 133 records in v1.5.0; those additions are outside this paper's IRR and model experiments. Table 3 keeps the experimental population and the reliability sample distinct.

Table 3: Experimental population and retained reliability sample.

Framework	Released records	Second annotation
PSM (§1910.119)	55	19
HazCom (§1910.1200)	16	11
Recordkeeping (Part 1904)	26	12
LOTO (§1910.147)	11	8
Total	108	50

The resource contains 103 obligations, three prohibitions, and two permissions. Its size is appropriate for inspecting a concrete knowledge architecture and tracing errors, but not for training a general regulatory model or estimating performance across all OSHA obligations. Selection covered operational topics within four frameworks rather than an exhaustive probability sample of the regulations. Record verification can establish that reviewed content has source support; it cannot determine the denominator of all duties that should have been represented.

4.2 Verification protocol and acceptance conditions

The source review used a fixed worksheet and codebook. For every record, the reviewer checked the official page and paragraph, excerpt fidelity, support for the restatement, granularity, applicability, workflow trigger, and audit evidence. AI prepared comparisons and proposed dispositions; the author inspected and confirmed them. This is completed human review with AI assistance, and is subject to confirmation bias rather than being an independent expert audit.

The original worksheet recorded 22 PASS, six PASS_WITH_NOTE, and 61 FAIL_NEEDS_ CORRECTION dispositions. The 19-row follow-up recorded 17 passes, one pass with a note, and one supporting-clause correction. These are review dispositions, not independent model error counts. Recurrent scope encodings account for many failures. Other findings concerned citations, incomplete conditions, excerpts, and combined duties. The release builder implements the recorded corrections; the presence of implemented_from_human_verification means a correction workflow was completed, not that legal error has become impossible.

Our proposed minimum protocol for a small, high-stakes release is full source review of the released population, explicit omission reporting, separate label-agreement measurement, and a traceable correction history. A record is eligible for the reviewed release only after each required review dimension is completed and material failures are corrected or excluded; notes must identify residual ambiguity. This is a design recommendation grounded in the case study, not a universally validated verification standard. A larger resource would need a justified risk-based review design and evidence about missed defects before reducing review coverage.

Table 4: Distinct verification layers and their evidential limits.

Layer	Completed check	What it cannot establish
Structure	Schema, enums, IDs, export consistency	Legal fidelity or regulatory completeness
Source support	108 author-reviewed source records	Independent expert agreement or all missing duties
Label consistency	50 paired human annotations; no adjudication	Correctness or reliability on new records
Model behavior	24 development scenarios; 240 calls	Independent deployment performance or saved labor

The layers test different properties. Structural validity cannot establish a legally faithful restatement, and agreement cannot establish truth when both annotators share a mistaken convention. Recorded review dispositions describe a correction process; they do not establish perfect coverage or that validation layers detect disjoint error sets. This study does not provide a controlled causal ablation of the verification pipeline.

4.3 Independent annotation design

The reliability study retains 50 IDs originally selected by seeded simple random sampling (seed 42) from the 89-record historical release. The plan was frozen before the second annotation, and those same IDs were regenerated with v1.2.0 text. This was not stratified sampling, and the 19 newly added records had zero probability of inclusion in this retained sample. Accordingly, the estimates describe the retained portion of the resource; they are not a random-sample estimate for all 108 records or for OSHA as a whole.

The first labels are the author-reviewed labels in the frozen release. A second human completed the three fields without the first-label key. The codebook defines risk by the direct consequence of violating the specific duty; evidence by the single most direct artifact or observable condition; and trigger by the main activating event or state. The ordinal risk field has four choices, evidence has 14, and trigger has 17. Only one value is scored in each field. Source verification and IRR are thus distinct: all 108 records were reviewed for support, while 50 records received 150 second-annotator labels to assess consistency.

The completed answers were frozen before discussion, with no adjudication or label replacement. One mistaken header in column eight was interpreted as workflow_trigger by its fixed position; all 50 values were present and valid. In Cohen's coefficient [12], agreement is p_o, the observed matching fraction, and p_e is the agreement expected from the two empirical category distributions:

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \qquad p_e = \sum_c p_1(c) p_2(c). \tag{2}$$

We report percentile intervals from 10,000 paired-row bootstrap resamples. After inspecting the results, we added explicitly post-hoc ordinal analyses with linear and quadratic disagreement weights $|a-b|$ and $|a-b|^2$, respectively [13]. They complement, rather than replace, the originally specified nominal κ.

5 Evaluation Procedure

5.1 Scenarios and comparison conditions

The development benchmark contains 24 synthetic operational scenarios with organization context and reference sets of requirements, triggers, evidence types, and citations. Its reference decisions comprise 19 answer cases, four missing-information cases, and one escalation case. AI drafted the scenarios, and the author confirmed their mappings with assistance from another AI tool before model execution. Both evaluated model families participated in the broader development workflow. The set is therefore exploratory and AI-assisted, with no independent held-out test partition.

Five conditions use the same scenario, permitted output vocabulary, decision policy, and model within each provider: (i) all 108 stored official excerpts; (ii) BM25 top-10 excerpts; (iii) MiniLM top-10 excerpts; (iv) BM25/MiniLM rank-fusion top-10 excerpts; and (v) the same hybrid candidate IDs serialized with RKM fields. The first condition is full context over the stored excerpt corpus, not all text of the four regulations. All retrieval conditions index the same curated requirement text plus official excerpt. The comparison therefore does not contrast RKM against a language model deprived of source information.

Algorithm 1: Evaluated retrieval-and-generation pipeline

```
Input: reviewed records R, scenario q, context c, condition s
1. Form query = q + newline + c.
2. Rank R by BM25, dense similarity, or RRF.
3. Select all R for full context; otherwise top 10 IDs.
4. Serialize source excerpts, or RKM fields for those IDs.
5. Insert q, c, decision policy, and context into the prompt.
6. Call the configured model with the common output schema.
7. Parse JSON; retain answer, prompt hash, usage, and latency.
8. Score exact sets and decision against frozen references.
```

The prototype performs one constrained model call after retrieval. It does not autonomously browse, collect workplace evidence, execute actions, or run an iterative planning loop. Its JSON contains requirement IDs, trigger and evidence sets, exact citation strings, a decision, a rationale, and any missing-information or escalation reason. The implemented RKM projection includes requirement text, mandatory status, type, risk, applicability, primary trigger and evidence, and clause, URL, excerpt, and source hash. It omits additional-trigger details and does not implement symbolic applicability inference.

5.2 Retrieval and reproducible inference

The lexical index lowercases text and tokenizes with [a-z0-9]+. BM25 uses k_1=1.5 and b=0.75. With term frequency f(t,d), document length |d|, average length $\bar{L}$, and document frequency df(t), its implementation is:

$$\mathrm{BM25}(q,d) = \sum_{t \in q} \mathrm{IDF}(t) \frac{f(t,d)(k_1+1)}{f(t,d) + k_1(1 - b + b|d|/\bar{L})}. \tag{3}$$

$$\mathrm{IDF}(t) = \ln\left[1 + \frac{N - \mathrm{df}(t) + 0.5}{\mathrm{df}(t) + 0.5}\right].$$

Dense retrieval uses all-MiniLM-L6-v2 in Sentence Transformers [9], normalized embeddings, and exact dot-product search over 108 records. No approximate-nearest-neighbor index or fine-tuning is used. Hybrid retrieval fuses the first 20 results from each component using reciprocal rank fusion [10], then returns ten:

$$\mathrm{RRF}(d) = \sum_{s\,:\,d \in \mathrm{top20}(s)} \frac{1}{60 + \mathrm{rank}_s(d)}. \tag{4}$$

Scores are sorted descending, with record ID as the tie-breaker. A post-hoc retrieval-only sensitivity analysis also evaluates intfloat/e5-base-v2 [11] and BM25/E5 fusion on the same frozen scenarios, using its required query and passage prefixes. These new rankings do not replace the candidate lists used in the original model experiment.

The logged inference experiment uses API model identifiers gpt-5.6-terra and claude-sonnet-5, with low reasoning effort, a 2,000-token output limit, zero-shot prompts, and no explicitly set temperature. Each of two providers runs five conditions on 24 scenarios, yielding 240 saved successful responses. The runner allows up to three attempts per call, 180-second request timeouts, and exponential backoff. Output is parsed as JSON and duplicate set entries are checked. Exact prompt reconstruction reproduced all 240 stored prompt hashes. Appendix A provides dependency versions, complete implementation steps, and archive entry points.

5.3 Metrics and interpretation

For a reference set G and a predicted set P, we score exact string-set precision, recall, and F1, then average F1 over scenarios. Both sets empty receive F1=1; exactly one empty receives zero. Decision accuracy is the fraction of exact matches to the three reference decision labels.

$$F_1(G, P) = \frac{2|G \cap P|}{|G| + |P|}, \qquad \text{macro } F_1 = \frac{1}{n}\sum_i F_1(G_i, P_i). \tag{5}$$

Retrieval reports Recall@10, reciprocal rank within ten, and binary-relevance nDCG@10. For the RKM-minus-hybrid contrast, paired bootstrap resampling uses scenarios as the unit, 10,000 resamples, and seed 20260927. Intervals quantify variation within these 24 cases; they do not remove scenario-construction bias, shared-regulation dependence, or model-family involvement. All comparisons are exploratory and intervals are not adjusted for multiple metrics. A single generation per model-condition-case does not measure run-to-run API variability.

Operational reference labels are derived from the same reviewed knowledge representation exposed to the RKM condition. This is intentional for testing recovery of stored operational annotations, but it favors access to those annotations. The contrast also adds curated paraphrases, applicability information, and context tokens together. It does not identify the isolated causal effect of JSON formatting, prove superior legal semantics, or establish performance on independently authored unseen rules. These boundaries are essential when interpreting gains in trigger and evidence F1.

6 Results and Failure Analysis

6.1 Annotation agreement

Table 5: Unadjudicated nominal agreement. Intervals use 10,000 paired-row bootstrap resamples.

Field	Matches	Agreement	Cohen κ	95% interval
Risk	23/50	46%	0.234	0.048–0.418
Evidence	34/50	68%	0.642	0.491–0.775
Trigger	29/50	58%	0.443	0.269–0.605

Evidence has the highest observed consistency, followed by workflow trigger; exact risk agreement is lower. The earlier target $\kappa \geq 0.70$ was not achieved as a nominal point estimate for any field. Larger label inventories and overlapping category boundaries may contribute to disagreement, but they do not establish that the scores are satisfactory by themselves, because κ already adjusts for chance agreement under the empirical marginals.

Risk assignments match in 23 cases, differ by one level in 24, and by two levels in three. Thus 47/50 are within one level. The second annotator assigns a higher risk in 25 of the 27 disagreements. Post-hoc linear weighted κ is 0.386 (95% bootstrap interval 0.215–0.542), and quadratic weighted κ is 0.547 (0.347–0.695). The weighting convention changes the apparent degree of agreement; none of these statistics validates the risk labels as calibrated probabilities of harm.

The disagreement pattern is informative for maintenance. Evidence differences often concern a broad written program versus a more specific record or artifact. Trigger differences include initial employment versus assignment to a process, and a continuous obligation versus an activating event. These are hypotheses about category boundaries, supported by the label pairs, rather than verified explanations of the annotator's reasoning. The second worksheet contained no explanatory notes. Preserving both label sets exposes these ambiguities for future codebook development.

6.2 Retrieval results

Table 6: Retrieval on 24 development scenarios. E5 rows are post-hoc sensitivity analyses.

Retriever	Recall@10	MRR@10	nDCG@10
BM25	0.804	0.847	0.754
MiniLM	0.801	0.765	0.691
BM25 + MiniLM (RRF)	0.817	0.835	0.749
E5-base-v2*	0.784	0.847	0.730
BM25 + E5 (RRF)*	0.808	0.858	0.753

Fusion of BM25 and MiniLM produces the highest observed recall, whereas BM25/E5 has the highest reciprocal rank. E5 alone does not outperform BM25 in recall on this small set. The result counters a keyword-only comparison while leaving substantial room for newer encoders, rerankers, larger corpora, and independent scenarios. No state-of-the-art retrieval claim is made.

6.3 Model outputs under five conditions

Table 7: Mean output scores on the development set. R/T/E/C are requirement, trigger, evidence, and citation F1; D is decision accuracy.

Model	Condition	R	T	E	C	D
GPT	All excerpts	0.881	0.690	0.606	0.857	0.917
GPT	BM25	0.825	0.710	0.716	0.750	0.958
GPT	MiniLM	0.800	0.707	0.677	0.763	0.917
GPT	Hybrid	0.801	0.693	0.681	0.805	0.917
GPT	RKM context	0.789	0.824	0.838	0.789	0.917
Claude	All excerpts	0.819	0.535	0.554	0.819	0.958
Claude	BM25	0.768	0.568	0.530	0.685	0.958
Claude	MiniLM	0.681	0.614	0.502	0.681	0.917
Claude	Hybrid	0.745	0.628	0.544	0.703	0.958
Claude	RKM context	0.771	0.858	0.888	0.757	0.958

Full-excerpt context achieves the highest requirement F1 for both models. Thus additional retrieval machinery and structured labels do not uniformly improve requirement discovery. Under identical hybrid candidates, RKM increases trigger F1 by 0.131 and evidence F1 by 0.157 for GPT-5.6-terra, and by 0.231 and 0.344 for Claude Sonnet 5. The paired intervals exclude zero for these four operational-label differences. Requirement and citation intervals include zero, and decision accuracy is unchanged. Figure 2 displays all five paired contrasts rather than only the positive ones.

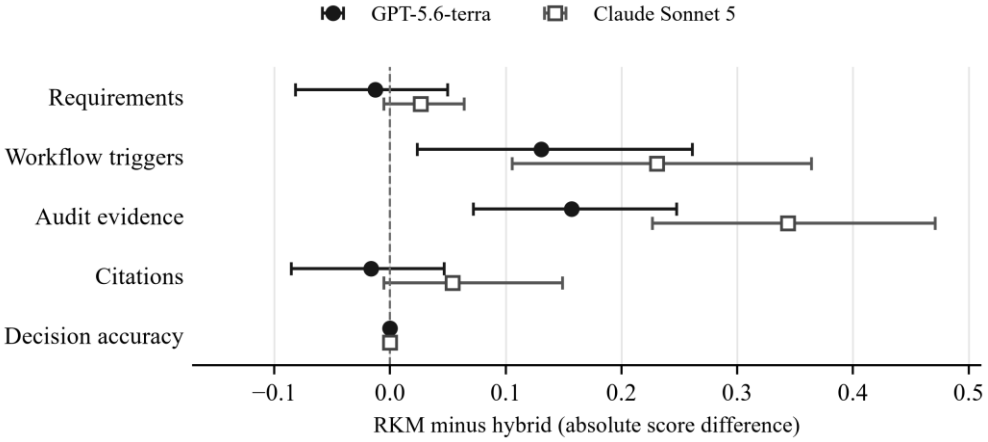


Figure 2: Paired RKM-minus-hybrid differences across all output metrics. Bars show 95% paired bootstrap intervals over 24 development scenarios; comparisons are exploratory.

RKM also increases context length: mean input tokens rise from 2,442 to 4,420 for GPT-5.6-terra and from 4,441 to 7,250 for Claude Sonnet 5 relative to hybrid. The measured gain is therefore the benefit of the complete curated-context package, with its additional information and cost. Across all five conditions, recorded usage at the runner's configured rates gives approximately USD 1.65 and USD 2.35, respectively; these are experiment cost estimates, not current provider price quotations.

A worked scored case illustrates the distinction between output organization and legal adequacy. In DEV-001, a plant introduces a hazardous solvent and decants it into workplace containers. The reference set contains four requirement IDs. GPT-5.6-terra's hybrid and RKM outputs each return five IDs, of which three overlap the reference, giving requirement F1=2×3/(4+5)=0.667 in both conditions. Their trigger sets recover one and two of three reference labels, yielding F1=0.500 and 0.800. This is an improvement in reference-label recovery without an improvement in requirement selection. Moreover, the frozen reference includes onboarding_new_employee even though the scenario concerns a new hazard for existing employees. This illustrates the limitation of aggregating primary record-level triggers into scenario labels; matching that convention is not proof that it is the best event-specific interpretation.

7 Human Review and Safety Implications

The practical value of an audit-support architecture is that a reviewer can locate the relevant duty, inspect the exact source, and see the proposed evidence category without reconstructing the whole chain from prose. Reuse across queries can make review more organized. We did not measure review time, real workplace evidence collection, or missed legal defects, so the experiment does not establish a reduction in human labor.

The recorded work comprises full author review of 108 records, confirmation of 24 scenario mappings, and 150 second-annotator labels on 50 records, in addition to follow-up version checks. These counts describe completed review units, not interchangeable units of labor. For a live compliance decision, someone still needs to verify site facts, conditions and exceptions, document authenticity, actual completion of the required action, and the sufficiency of evidence. AI-prepared source comparisons and checklists assist those tasks without removing the need for them.

Table 8: Post-hoc review-routing audit for the RKM condition. Exact answer match requires all four reference sets and the decision to match.

Model	Answer	Other route	Exact answer match
GPT-5.6-terra	17/24	7/24	5/17
Claude Sonnet 5	18/24	6/24	5/18

The RKM outputs route seven of 24 scenarios to missing information or escalation for GPT-5.6-terra and six for Claude Sonnet 5. Yet only five answer-routed cases per model exactly match all four reference sets as well as the reference decision. The remaining mismatches are not all proven legal errors, because the references themselves encode imperfect conventions; they nevertheless show why routing accuracy cannot serve as a safety guarantee. All five reference cases requiring more information or escalation receive a non-answer decision, but one escalation example and four incomplete-fact examples are far too few to estimate sensitivity to realistic out-of-distribution events.

To study human–AI verification allocation, let z_i=1 mean an output requires substantive human review, c_i be its measured review cost, and L_i be the loss from accepting an unreviewed output. A future policy could minimize total review cost subject to a validated limit on residual loss:

$$\min_{\mathbf{z}} \sum_i c_i z_i, \qquad \text{subject to } \mathbb{E}\left[\sum_i (1 - z_i) L_i\right] \leq \varepsilon. \tag{6}$$

Equation 6 states a prospective evaluation objective; the current system neither estimates L_i nor solves this optimization. Deployment would require independent calibration data, realistic miss costs, measured human error and time, and a policy for unknown conditions. The present categories are routing instructions, not confidence estimates. A high predicted confidence, a valid hash, or a low stored risk category cannot authorize bypassing qualified review.

RKM is structurally applicable beyond OSHA: stable duty identifiers, source references, operational triggers, and evidence categories are not tied to a particular regulator. The populated vocabulary, industry descriptors, examples, and empirical results are OSHA-specific. Transferring the architecture requires new source adapters, domain-specific applicability and exception rules, and independent verification; cross-domain validity has not been demonstrated.

8 Limitations and Reproducibility

This case study has three distinct sample sizes: 108 curated records, 50 retained IDs for annotation agreement, and 24 development scenarios for model comparisons. None is enlarged by the number of model calls. They support an inspectable implementation and descriptive comparisons, but not broad generalization, a definitive sample-size adequacy claim, or certification of high-stakes automation. The retained IRR sample excludes the 19 new records; the benchmark has no independent holdout, and its operational references share the represented taxonomy. Source verification was author-led and AI-assisted rather than an external regulatory audit.

The schema remains an approximation. Primary labels compress multiple possible triggers or artifacts, applicability descriptors omit some legal conditions, and risk categories are judgments rather than validated severity measurements. Source-linked snapshots do not implement a temporal legal graph. The system does not inspect real audit evidence, assess adversarial behavior, or test prompt-injection resistance. Those limitations restrict the safety claim to traceability, constrained output, and an explicit review interface.

Reproducibility is strongest for the retained evaluation: code, scenario inputs, rankings, model requests reconstructible from the runner, saved responses, and scoring are available in the accompanying anonymous research archive. The archive also includes original and supplementary IRR calculations and the post-hoc E5 analysis. Identifying curator fields and local paths are replaced only in distribution copies, with a manifest documenting the transformation; scientific labels and model answers are preserved. Historical extraction can be inspected through source and correction artifacts but cannot be rerun exactly because complete historical prompts and parameters were not retained. Provider aliases may also change, so reproducing saved-output scores is more exact than regenerating API answers.

Anonymous supplementary artifacts accompany this submission, including the implementation and retained evaluation data. Author attribution and public repository details are reserved for a non-anonymous release.

9 Conclusion

ComplianceKB demonstrates a concrete method for turning selected regulatory provisions into maintained, source-linked obligations with operational audit fields. Its contribution lies in making the interpretation, validation, and review boundary inspectable. Full source review corrected defects that

structural tests could not establish, while the independent second annotation exposed remaining taxonomy ambiguity. In a controlled development comparison, the complete RKM context improves recovery of trigger and evidence labels, but it does not consistently improve requirement selection, citations, or decisions. These results support a reusable audit-support method and an explicit research agenda for human review allocation, with clear limits on autonomy and generalization.

AI Use Disclosure

Claude-family and OpenAI tools assisted extraction drafts, coding, verification worksheets, scenario construction, analysis, and manuscript editing. Human reviewers were responsible for the recorded source confirmations and annotation decisions. The historical log does not identify every extraction run precisely; the inference study instead retains the exact API identifiers and settings reported in Section 5. AI assistance in constructing and checking the scenario references is a source of evaluation dependence and is disclosed as such. Neither AI tool is counted as a human IRR annotator.

References

[1] M. Hashmi, G. Governatori, H.-P. Lam, and M. T. Wynn. Are we done with business process compliance: state of the art and challenges ahead. Knowledge and Information Systems 57:79–133, 2018. doi:10.1007/s10115-017-1142-1.

[2] OASIS. LegalRuleML Core Specification Version 1.0. OASIS Standard, 2021. https://docs.oasis-open.org/legalruleml/legalruleml-core-spec/v1.0/os/legalruleml-core-spec-v1.0-os.html.

[3] H. Hettiarachchi et al. CODE-ACCORD: A Corpus of building regulatory data for rule generation towards automatic compliance checking. Scientific Data 12:170, 2025. doi:10.1038/s41597-024-04320-x.

[4] Y. Koreeda and C. D. Manning. ContractNLI: A Dataset for Document-level Natural Language Inference for Contracts. Findings of EMNLP, pp. 1907–1919, 2021. doi:10.18653/v1/2021.findings-emnlp.164.

[5] J. Sun, Z. Luo, and Y. Li. A Compliance Checking Framework Based on Retrieval Augmented Generation. COLING, pp. 2603–2615, 2025. https://aclanthology.org/2025.coling-main.178/.

[6] T. Gokhan and T. Briscoe. Grounded Answers from Multi-Passage Regulations: Learning-to-Rank for Regulatory RAG. Natural Legal Language Processing Workshop, pp. 135–146, 2025. doi:10.18653/v1/2025.nllp-1.10.

[7] T. Gokhan, K. Wang, I. Gurevych, and T. Briscoe. Shared Task RIRAG-2025: Regulatory Information Retrieval and Answer Generation. RegNLP Workshop, pp. 1–4, 2025. https://aclanthology.org/2025.regnlp-1.1/.

[8] R. El-Yaniv and Y. Wiener. On the Foundations of Noise-free Selective Classification. Journal of Machine Learning Research 11:1605–1641, 2010. https://www.jmlr.org/papers/v11/el-yaniv10a.html.

[9] N. Reimers and I. Gurevych. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. EMNLP-IJCNLP, pp. 3982–3992, 2019. doi:10.18653/v1/D19-1410.

[10] G. V. Cormack, C. L. A. Clarke, and S. Buettcher. Reciprocal rank fusion outperforms Condorcet and individual rank learning methods. SIGIR, pp. 758–759, 2009. doi:10.1145/1571941.1572114.

[11] L. Wang et al. Text Embeddings by Weakly-Supervised Contrastive Pre-training. arXiv:2212.03533, 2022. Model technical reference (preprint). https://arxiv.org/abs/2212.03533.

[12] J. Cohen. A coefficient of agreement for nominal scales. Educational and Psychological Measurement 20(1):37–46, 1960. doi:10.1177/001316446002000104.

[13] J. Cohen. Weighted kappa: nominal scale agreement with provision for scaled disagreement or partial credit. Psychological Bulletin 70(4):213–220, 1968. doi:10.1037/h0026256.

[14] Occupational Safety and Health Administration. 29 CFR 1910.1200: Hazard communication. Official regulatory reference. https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.1200. Accessed September 30, 2026; experimental source version stated separately in Section 3.3.

[15] Occupational Safety and Health Administration. 29 CFR 1910.119: Process safety management of highly hazardous chemicals. Official regulatory reference. https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.119. Accessed September 30, 2026; experimental source version stated separately in Section 3.3.

Appendix A Implementation and Reproduction Details

Table 9: Implementation configuration and reproducibility entry points.

Component	Setting or artifact
Frozen inputs	data/releases/v1.2.0/all_requirements.json; 108 records. Confirmed development scenarios: 24 rows.
Retrieval runtime	Python 3.12.14; PyTorch 2.14.0+cpu; Sentence Transformers 6.1.0; Transformers 5.17.0; NumPy 2.3.5.
MiniLM	all-MiniLM-L6-v2; L2 normalization; exact dot product. Original immutable weight revision was not logged.
E5 sensitivity	intfloat/e5-base-v2; revision f52bf8ec8c7124536f0efb74aca902b2995e5bcd; query:/passage: prefixes; 512 tokens; CPU; seed 20260930.
API generation	gpt-5.6-terra / claude-sonnet-5; reasoning low; output limit 2,000; no temperature set; four workers. Python standard-library HTTP requests.
Prompt and output	run_llm_benchmark.py: build_context, build_prompt, answer_schema, provider calls. Saved JSON retains answers, prompt hashes, model metadata, usage and latency.
Scoring	evaluate_benchmark.py and aggregate_llm_results.py; set matching and paired bootstrap, without a model judge.
Independent annotation	irr_annotation_sample_COMPLETED.csv and irr_annotator1_key.csv; finalize_irr_unadjudicated.py preserves raw labels.
Revision checks	revision_evidence_audit.py verifies 240 prompt hashes and calculates supplementary weighted κ and routing counts.
Distribution	Anonymous archive manifest records source/distribution checksums and sanitization. API keys and model weights are excluded.

Algorithm 1 maps directly to build_context and build_prompt in run_llm_benchmark.py. The system instruction requests source-grounded regulatory reasoning and conformance to a JSON schema. The user prompt contains the scenario, organization context, task names, decision policy, and condition-specific context. Scenario gold columns are not inserted into the prompt. All 108 IDs are permitted by the response schema, so an enum-valid ID is not by itself proof that it was supported by the supplied context. The scoring script compares predicted sets to references without an LLM judge. Free-text rationale truthfulness was not independently scored.

The retrieval-only runner's earlier scenario file differs from the confirmed file in review metadata, not scenario text, organization context, or relevant-ID sets. We verified this equivalence before reusing its rankings. A separate revision audit reconstructs all 240 prompts and checks their original hashes. The anonymous release must therefore preserve every field that enters a prompt, even when removing names from curation metadata.

```
python research/benchmark/aggregate_llm_results.py
python research/notebooks/revision_evidence_audit.py
python research/notebooks/finalize_irr_unadjudicated.py
# Additional weight download; no hosted LLM calls:
python research/benchmark/revision_retrieval_sensitivity.py
```

The first command recomputes saved-output metrics without an API key. The second checks prompt reconstruction, source-file relationships, decision routing, and the additional ordinal IRR analysis. Re-executing hosted models requires the user's own provider credentials and incurs new API costs; credentials are neither stored nor included. Model-weight downloads are separate from the archive. The E5 sensitivity script pins the snapshot revision recorded in its manifest; the original MiniLM run logged the model name but not its immutable weight revision. This remaining limitation is retained rather than inferred from a current download.

Appendix B Reporting Boundary for Audit Support

Table 10: Proposed reporting checklist separating automation from human responsibility.

Question	Reported evidence	Human responsibility / remaining study
What was represented?	Selected duties and source clauses	Coverage denominator, omissions, exceptions
Is a record traceable?	Version, excerpt, normalized hash	Authority, date relevance, legal interpretation
Are labels reproducible?	Two label sets and raw agreement	Ambiguous taxonomy and adjudication policy
Does retrieval recover duties?	Rankings, relevance sets, Recall@10	Independent scenario and relevance construction
Does the answer help an audit?	Requirement, trigger, evidence and citation scores	Actual evidence quality and workplace facts
Can review be reduced?	Routing counts only in this study	Measure review cost, residual loss and calibration

For future replication, the minimum useful report should separate verified-record count, source-version scope, omission coverage, independent annotation design, retrieval configuration, reference construction, output errors, and remaining human work. This reporting proposal follows the observed failure modes; it has not been established by consensus or validated across regulatory domains. It is intended to make future systems comparable on the claims they actually test.